

Q: 2(i)

A: 2(i)

* Medicinal Chemistry :-

Studies the interaction b/w drugs and biological targets, and the development of new medicinal targets.

* Polymer chemistry :-

Studies polymers, their types, their behaviour, their structure and uses.
for examples - nylon, teflon coated bags.

Q: 2(ii)

A: 2(ii)

A dilute solution contains a relatively small amount of dissolved solute, while a concentrated solution contains a relatively large amount of solute in the same amount of solvent. Brine is a concentrated solution because it holds a large amount of dissolved NaCl.

Q: 2(iii)(or)

A: 2(iii)(or)

Max electrons in sub-shells : $s = 2$, $p = 6$, $d = 10$, $f = 14$ electrons max. The M-shell ($n=3$) has $3s^2$, $3p^6$, $3d^{10}$ sub-shells, so its maximum capacity = $2 + 6 + 10 = 18$ electrons.

Q: 2(iv)(or)

A: 2(iv)(or)

Unknown element: $3s^1 \dots \dots n=3 \rightarrow \text{period} = 3$
1 valence electron $\rightarrow$ group = 1, readily loses this electron, so it is a metal (this is sodium).

Q: 2(v)

A: 2(v)

- ① Iron : Haber process (ammonia synthesis).
- ② Platinum/Palladium : Catalytic converters.
- ③ Nickel : manufacture of magazine
- ④ Platinum/vanadium pentoxide : Contact process (sulphuric acid).

Q: 2(vi)

A: 2(vi)

BF_3 has only 3 bonding pairs around Boron and is electron-deficient. Ammonia's nitrogen has a lone pair which it donates entirely to boron forming a coordinate (dative) covalent bond, represented $\text{H}_3\text{N} \rightarrow \text{BF}_3$.

Q: 2(vii)

A: 2(vii)

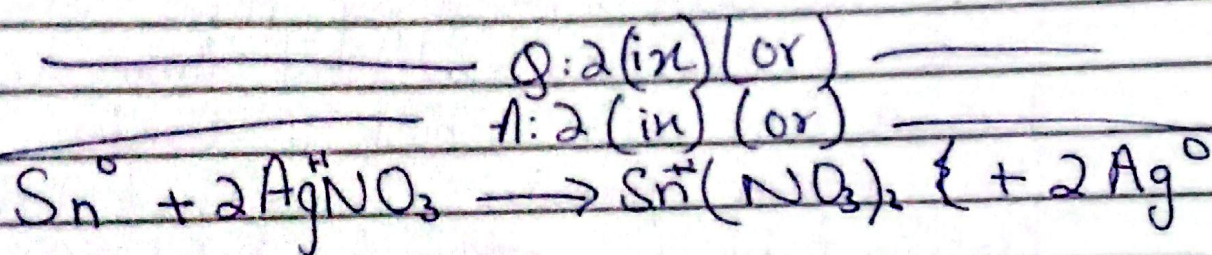
- ① Refractory bricks (chemical inertness, heat stability).
- ② Electrodes / Anodes in ~~carbon brushes~~ in electric motors.
- ③ Moderator rods in nuclear reactors (inertness), carbon brushes in electric motors.

O: 2 (viii)
A: 2 (viii)

Q Molar mass of KMnO_4 :
 $39 + 55 + 4(16)$
 158 g/mol

Q Moles : $\frac{\text{mass}}{\text{M.m}}$

$$= \frac{31.6}{158} \Rightarrow 0.2 \text{ mol.}$$



$\text{Sn} = 0 \rightarrow +2$ (oxidized) (reducing agent)
 $\text{Ag} = +1 \rightarrow 0$ (reduced) (oxidizing agent)

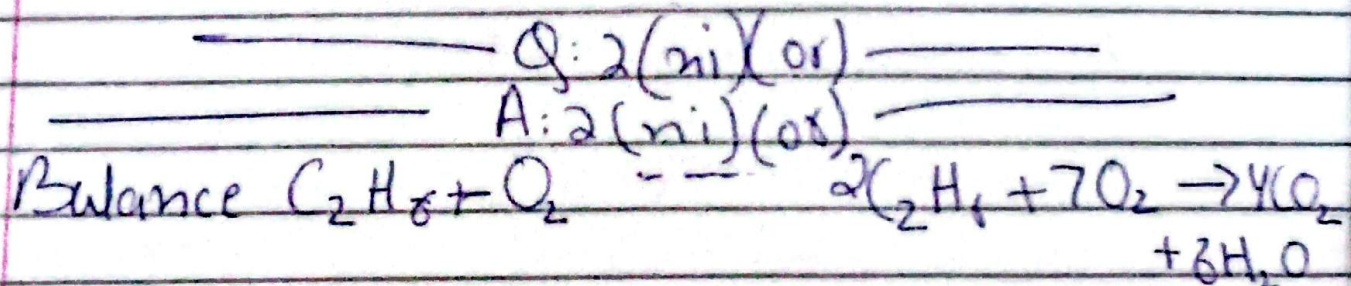
O: 2 (n)
A: 2 (n)

The study of energy changes is called in reaction is chemical energetics.

a reaction releasing heat is exothermic

a reaction absorbing heat is endothermic

ΔH is negative for exothermic reaction and ΔH is positive for endothermic reaction, bond breaking always absorbs energy, bond releasing always releases energy, a catalyst lowers activation energy without changing the overall enthalpy of reaction.



$$\text{C: } 4 = 4$$

$$\text{H: } 12 = 12$$

$$\text{O: } 14 = 14$$

$$\begin{array}{c} \text{Q: } 3(\text{li}) \\ \text{A: } 3(\text{li}) \end{array}$$

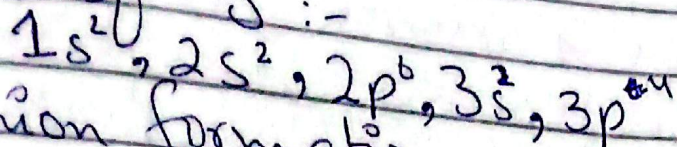
@ total = $\text{RAM} \times 100$
 $= 35.5 \times 100$
 $= 3550$

contribution of A = 75.77×35
 2651.95

contribution of B: $3550 - 2651.95$
 $= 898.05$

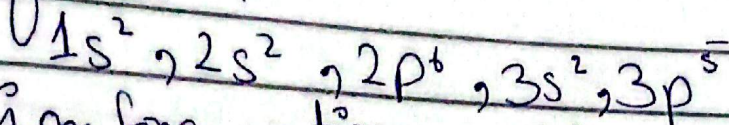
Isotope of B : $898.05 \div 24.23$
 $\approx 37.06 \text{ amu}$

E.C of S :-

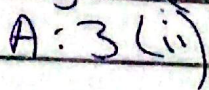
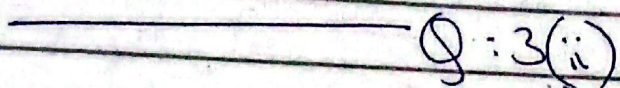


Anion formation :- -2

E.C of Cl :-



Anion formation :- -1



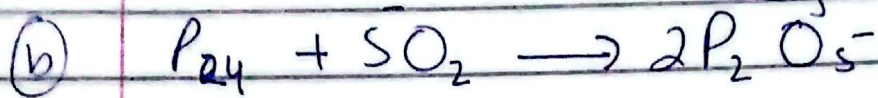
a) $C = \frac{85.7}{12} \Rightarrow \text{BT} : 14$

$H : \frac{14.3}{1} = 14.3$

ratio : $1 : 2$

Empirical CH_2 (mass 14) $n = \frac{56}{14} \Rightarrow 4$

molecular C_4H_8

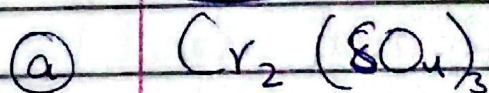


$$P = 4 : 4$$

$$O_2 = 10 = 10$$

~~Q.3 (iii)~~

~~A: 3 (iii)~~



~~2(n) + 3(-2) = 0~~
~~2n + 3(-2) = 0~~
~~2n + 3(-2) = 0~~
~~2n = 6~~

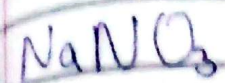
$$2(n) + 3(SO_4^{2-}) = 0$$

$$2(n) + 3(-2) = 0$$

$$2n - 6 = 0$$

$$2n = 6$$

$$n = \frac{6}{2} \quad n = 3$$



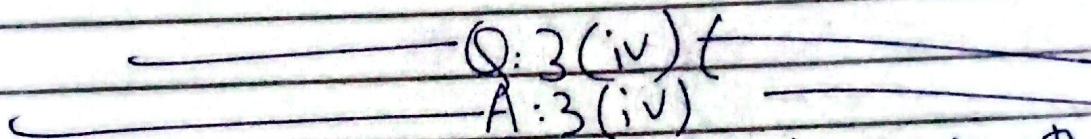
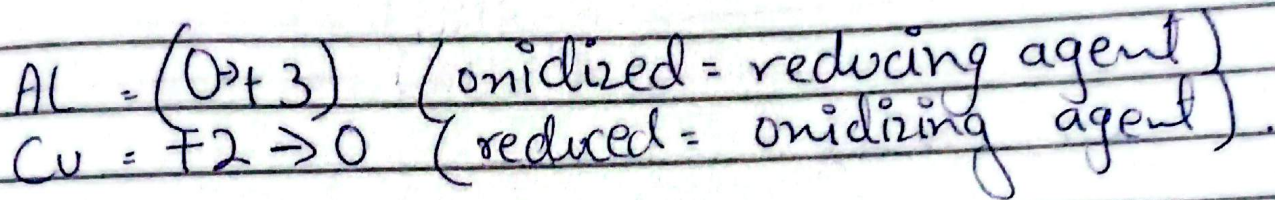
$$+1 + n + 3(-2) = 0$$

$$1 + n - 6 = 0$$

$$n - 6 + 1 = 0$$

$$n - 5 = 0$$

$$n = 5$$



Q In diamond, each C bonds to 4 others in a rigid 3-D tetrahedral network with no free electrons $\rightarrow$ extremely hard, electrical insulator. In graphite, each C bonds to only 3 others in 2D hexagonal layers, the 4 delocalized electron moves freely within each layer (conducts electricity), and weak van der Waals forces b/w layers let them slide $\rightarrow$ soft, slippery conducting.

Q Diamond: Cutting/drilling tools (hardness)?

medical tools e.g. cataract surgery (hardness / precision).

Graphite : Electrodes in furnaces (conducts + heat resistant) ; Pencils / lubricant (soft + layered).

MCQs:-

1. A
2. D
3. B
4. C
5. B
6. C
7. C
8. A
9. C
10. B
11. B
12. C